SAM THEIL

MECHANICAL ENGINEERING AT THE UNIVERSITY OF CINCINNATI

CONTACT

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Cincinnati Ohio

SKILLS

Design

- **Fusion 360**, **Inventor**, used on various projects involving **GD&T**, **FEA Analysis**, designing for CNC and additive manufacturing
- **Autodesk Ultimate CFD**

Mechanical

- **3D Printing** and **Laser Cutting** for rapidly prototype parts
- Manual **CNC machining** for test parts
- **Statistical tolerance analysis** used on tight tolerance parts

Electrical and software

- **Python**, **MATLAB**, **GitHub**
- Front-end and back-end **web development**
- Agile Framework
- **Soldering**
- **Wiring**

EDUCATION

University of Cincinnati

- As a high school student, completed 57 college credit hours in science & engineering related courses.
- 3.93 GPA
- Officially starting college in Fall 2024

Theil Academy Homeschool

- Will recieve high-school diploma in June 2024
- Weighted GPA: 4.49

PROFILE

A passionate and technically adept leader with strong people skills and a diverse skill set in engineering, entrepreneurship, and design. Demonstrated excellence in mechanical engineering, rocketry, and 3D printing, coupled with a creative mindset. Balances technical expertise with interpersonal skills, growing leadership qualities, and a commitment to personal and professional growth.

PROFESSIONAL EXPERIENCE

eRockets | CAD Designer 2021 - 2023

- Developed a Fusion 360 plugin for automatically designing model rocket nose cones of different shapes and dimensions. Plugin is still in use today.
- Managed 3D print farm operations with other employees; optimized design and tolerances for nose cones.

Contract Web Developer 2023 - Present

- Built web app using low-code tools that utilized OpenAI API (ChatGPT) to automate the writing of user requirements for software development (Epics, Features, and Stories in the Agile framework).
- Successful launch and positive customer response to the web app helped secure $2.5 million in funding for my client.

PROJECTS

Rocket Airbrakes 2023 - 2024

- Designed, manufactured, and programmed a rocket airbrake system that controlled the rocket's peak altitude
- Wrote software to continuously calculate the current state (drag, projected altitude, etc) adjusting brakes during flight to optimize for a target altitude.
- Developed simulation and graphing in Python that allowed for troubleshooting and quick tuning of the system.
- Developed many optimizations including implementing a Kalman filter on key data to improve state estimation, and incorporating orientation data to more accurately predict altitude, etc.
- Allowed my rocket team to win 9th in the largest rocketry competition in the US

High Powered Level 1 Rocket 2022 - 2023

- Modeled and built a fully parametric 3D printed high power rocket. Later used the parametric capability to easily build a second, slightly smaller rocket.

Automatic Dog Feeder 2023 - 2024

- Fully autonomous dog feeder with graphical web interface for adjusting time of meal and on-demand treat dispensing.

TEAMWORK AND LEADERSHIP

American Rocketry Challenge 2020 - 2024

Team captain and leader from 2021-2024

- Led team formation, recruiting, and fundraising.
- Encouraged and challenged the team to hit commitments
- Used Agile processes to guide team through the design and build process
- Implemented SOP for team launch processes
- Guided and taught new team members rocketry basics
- Together, we built 25+ mid-power model rockets and launched 226 times